Lecture 2

- We saw that the general solution of a linear ODE can be expressed in terms of complementary and particular solutions.
  - Complementary solutions form a solution space.
  - For 1st-order ODEs the solution space is 1D and for 2nd-order ODEs they are 2D ... more on this today
- We learnt how to use reduction of order to solve a 2nd-order ODE when one complementary solution is known

Lecture 3

We will:

1. Define linear independence of functions and discuss the importance of linear independence of complementary solutions

2. Learn how to use the Wronskian to determine linear independence of a set of functions

3. Connect linear independence with trial solutions

Existence and uniqueness of solutions

Theorem

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$$

If f_0, f_1 and g are all continuous over $D = [a, b]$, then over D:

- ODEs: The general solution to $\mathsf{T}y = g$ is a linear combination of 2 linearly independent complementary solutions and a single particular solution.
  - Solutions space (i.e., all complementary solutions) is a 2-dimensional space.
- IVPs: When supplemented with initial values $y(t_0)$ and $y'(t_0)$ where $t_0 \in D$, it has a unique solution which is twice differentiable.

Linear independence

Linear algebra:

Two non-zero vectors, $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, are linearly dependent if there exists c such that $\boldsymbol{v}_1 = c\,\boldsymbol{v}_2$.

Differential equations:

Two non-zero functions, $y_1(t)$ and $y_2(t)$, are linearly dependent if there exists c such that

$$y_1(t) = c\,y_2(t) \quad \text{for all} \quad t\,.$$

If not, then they are linearly independent.

Exercise 1.6: Are $y_1 = \mathrm{e}^{-2t}$ and $y_2 = 2\mathrm{e}^{-2t}$ linearly independent?

No, because $y_1 = \dfrac{1}{2} y_2$

The Wronskian

Definition (The Wronskian)

Given two functions y_1 and y_2, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2](t) = \det \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

- y_1 and y_2 are linearly independent if

$$\mathrm{wr}[y_1, y_2](t) \neq 0 \quad \text{for some } t$$

  - y_1 and y_2 are linearly dependent if $\mathrm{wr}[y_1, y_2](t) = 0$ for all t
  - $\mathrm{wr}[y_1, y_2] = 0$ for some t tells you nothing

Theorem (Abel's differential equation identity)

Let y_1 and y_2 be two solutions of $y'' + f_1 y' + f_0 y = 0$. Then

$$\mathrm{wr}[y_1, y_2](t) = \mathrm{wr}[y_1, y_2](t_0) \mathrm{e}^{-\int_{t_0}^{t} f_1(\tau) \mathrm{d}\tau}$$

- If the Wronskian is non-zero anywhere it will be non-zero everywhere.
- If the Wronskian is zero anywhere it will be zero everywhere.

If we know one solution, say y_1, we can use Abel's identity and Slide 5 to find the second linearly independent solution, y_2, by solving the ODE

$$y_1 y_2' - y_1' y_2 = \mathrm{wr}[y_1, y_2](t_0) \mathrm{e}^{-\int_{t_0}^{t} f_1(\tau) \mathrm{d}\tau}$$

This is equivalent to using reduction of order.

Exercise 1.7: Study linear independence of solutions to below ODE

$$y'' + by' + \frac{b^2}{4}y = 0 \quad \Rightarrow \quad y(t) = C_1\, \mathrm{e}^{-\frac{b}{2}t} + C_2\, t\mathrm{e}^{-\frac{b}{2}t}$$

Therefore, study Wronskian of $\mathrm{e}^{-\frac{b}{2}t}$ and $t\,\mathrm{e}^{-\frac{b}{2}t}$, which is

$$\begin{aligned}
\mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](t) &= \det \begin{pmatrix} \mathrm{e}^{-\frac{b}{2}t} & t\mathrm{e}^{-\frac{b}{2}t} \\ -\frac{b}{2}\mathrm{e}^{-\frac{b}{2}t} & \left(1 - \frac{b}{2}t\right)\mathrm{e}^{-\frac{b}{2}t} \end{pmatrix} \\
&= \left(1 - \frac{b}{2}t - t\left(-\frac{b}{2}\right)\right)\mathrm{e}^{-\frac{b}{2}t}\mathrm{e}^{-\frac{b}{2}t} \\
&= \mathrm{e}^{-bt}
\end{aligned}$$

which is non-zero for some t, e.g., $\mathrm{wr}(1) = \mathrm{e}^{-b}$.

Hence, these two solutions are linearly independent and are a basis for the solution space.

- Choose $t_0 = 0$
- Then $\mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](t_0) = \mathrm{e}^{-bt_0} = 1$
- Note $f_1 = b \Rightarrow \mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau} = \mathrm{e}^{-\int_0^t b\mathrm{d}\tau} = \mathrm{e}^{-bt}$
- Thus, $\mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](t) = \mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](0)\mathrm{e}^{-\int_0^t f_1(\tau)\mathrm{d}\tau} = 1 \times \mathrm{e}^{-bt} = \mathrm{e}^{-bt}$
- Confirming Abel's identity

Exercise 1.8: Find solutions of the form $\phi = r^\lambda$ for the below ODE

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}\phi}{\mathrm{d}r} - \frac{n^2}{r^2}\phi = 0\,, \qquad r > 0\,, \qquad n \in \{0, 1, 2, \ldots\}^*$$

We "trial"

$$\phi = r^\lambda \quad \Rightarrow \quad \phi' = \lambda\, r^{\lambda-1} \quad \Rightarrow \quad \phi'' = \lambda(\lambda-1)\, r^{\lambda-2}$$

Substituting into the ODE

$$\lambda(\lambda-1)r^{\lambda-2} + \frac{1}{r}\lambda r^{\lambda-1} - \frac{n^2}{r^2}r^\lambda = 0$$

$$\lambda^2 - n^2 = 0$$

which has solution $\lambda = \pm n$.

*This is the Euler equation from, e.g., electrostatics, which we will see again in Topic 5

For $n \neq 0$:

The Wronskian shows r^n and r^{-n} are linearly independent (details left as exercise).

Hence the general solution is $\phi = C_1 r^n + C_2 r^{-n}$.

For $n = 0$:

We have only one solution $\phi_1 = \mathrm{e}^{\pm 0} = 1$. (Our ansatz was not expressive enough.)

The Wronskian is proportional to $\mathrm{e}^{-\int \frac{1}{r}\mathrm{d}r} = \frac{1}{r}$

Thus, from Abel's identity

$$\phi_2' = \frac{1}{r} \qquad \Rightarrow \qquad \phi_2 = \log r.$$

so the general solution for $n = 0$ is $\phi = C_1 + C_2 \log r$

Higher-order (linear) ODEs

Theorem (Solutions of an nth order linear ODE/IVP)

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^n}{\mathrm{d}t^n} + f_{n-1}(t)\frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}} + \ldots + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t).$$

If $f_0,\ f_1, \ldots, f_{n-1}$ and g are all continuous over $D = [a, b]$, then over D:

- ODEs: The general solution to $\mathsf{T}y = g$ is the sum of n linearly independent complementary solutions and a single particular solution.
  - Solution space (i.e., all complementary solutions) form an n-dimensional vector space.
- IVPs: When supplemented with initial values $y(t_0), y'(t_0), \ldots, y^{(n-1)}(t_0)$ where $t_0 \in D$, it has a unique solution which is n-times differentiable.

Definition (The Wronskian)

Given n functions $y_1, y_2, \ldots, y_n$, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = \det \begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}$$

- $y_1, y_2, \ldots, y_n$ are linearly independent if there exists a t such that

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) \neq 0$$

  - $y_1, y_2, \ldots, y_n$ are linearly dependent if $\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = 0$ for all t

Method of trial solutions (higher order)

Exercise 1.9: Find the general solution to the below ODE by trialing $y = \mathrm{e}^{\lambda t}$.

$$y''' + y'' - 4y' - 4y = 0$$

$y = \mathrm{e}^{\lambda t} \Rightarrow (\lambda + 1)(\lambda^2 - 4) = 0$. Thus $\lambda = -1, \pm 2$

Let $y_1 = \mathrm{e}^{-2t}$, $y_2 = \mathrm{e}^{-t}$, $y_3 = \mathrm{e}^{2t}$. The Wronskian is

$$\begin{aligned} \mathrm{wr}[y_1, y_2, y_3](t) &= \det \begin{pmatrix} \mathrm{e}^{-2t} & \mathrm{e}^{-t} & \mathrm{e}^{2t} \\ -2\mathrm{e}^{-2t} & -\mathrm{e}^{-t} & 2\mathrm{e}^{2t} \\ 4\mathrm{e}^{-2t} & \mathrm{e}^{-t} & 4\mathrm{e}^{2t} \end{pmatrix} \\ &= 12\mathrm{e}^{-t} \neq 0 \end{aligned}$$

and so the general solution is $y = C_1\mathrm{e}^{-2t} + C_2\mathrm{e}^{-t} + C_3\mathrm{e}^{2t}$.

Reduction of order also works in higher dimensions but we won't cover that in MAST20030.

Summary of Linear ODEs

1. An nth order linear ODE has n *linearly independent* complementary solutions and a single particular solution. It has a unique solution if n initial values are specified.

2. Check for linear independence using the Wronskian.

3. If one complementary solution for a 2nd order ODE is known, the general solution can be found through reduction of order.